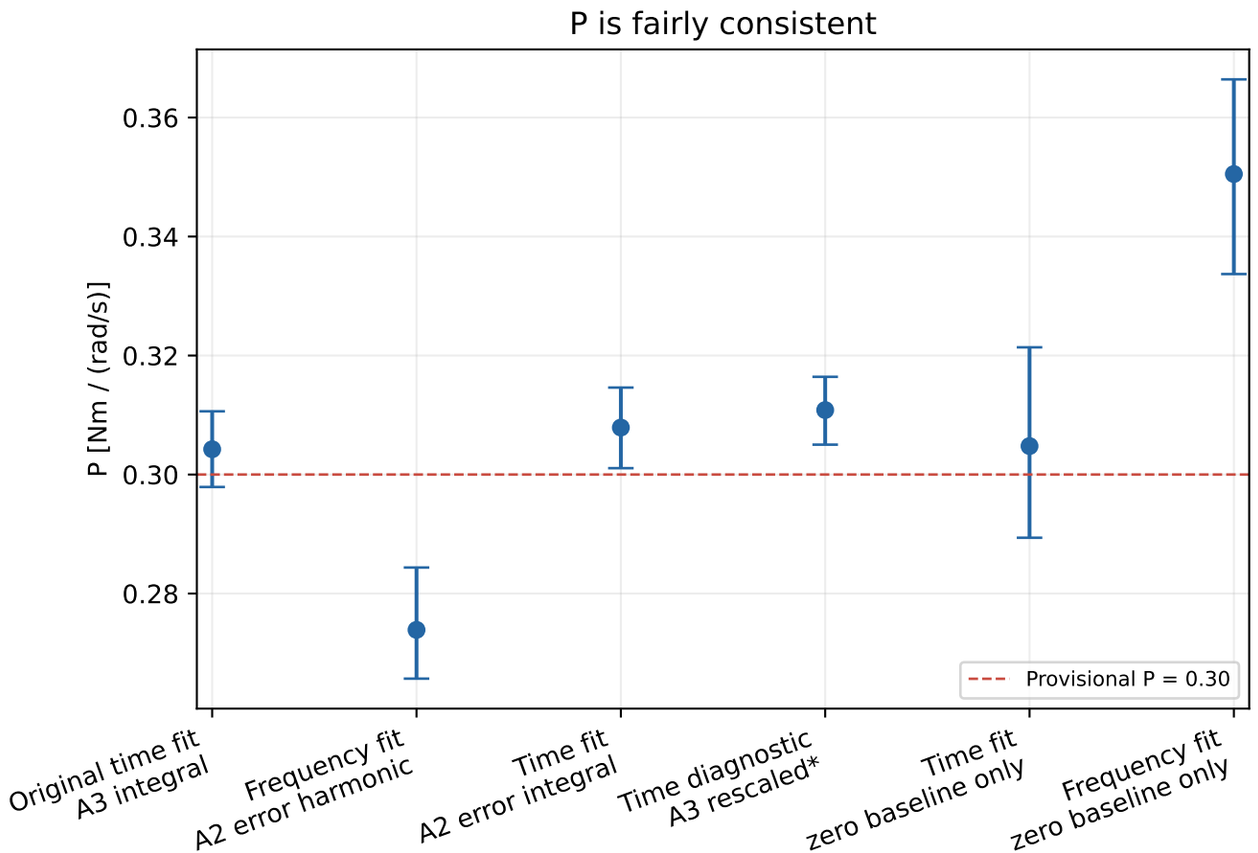
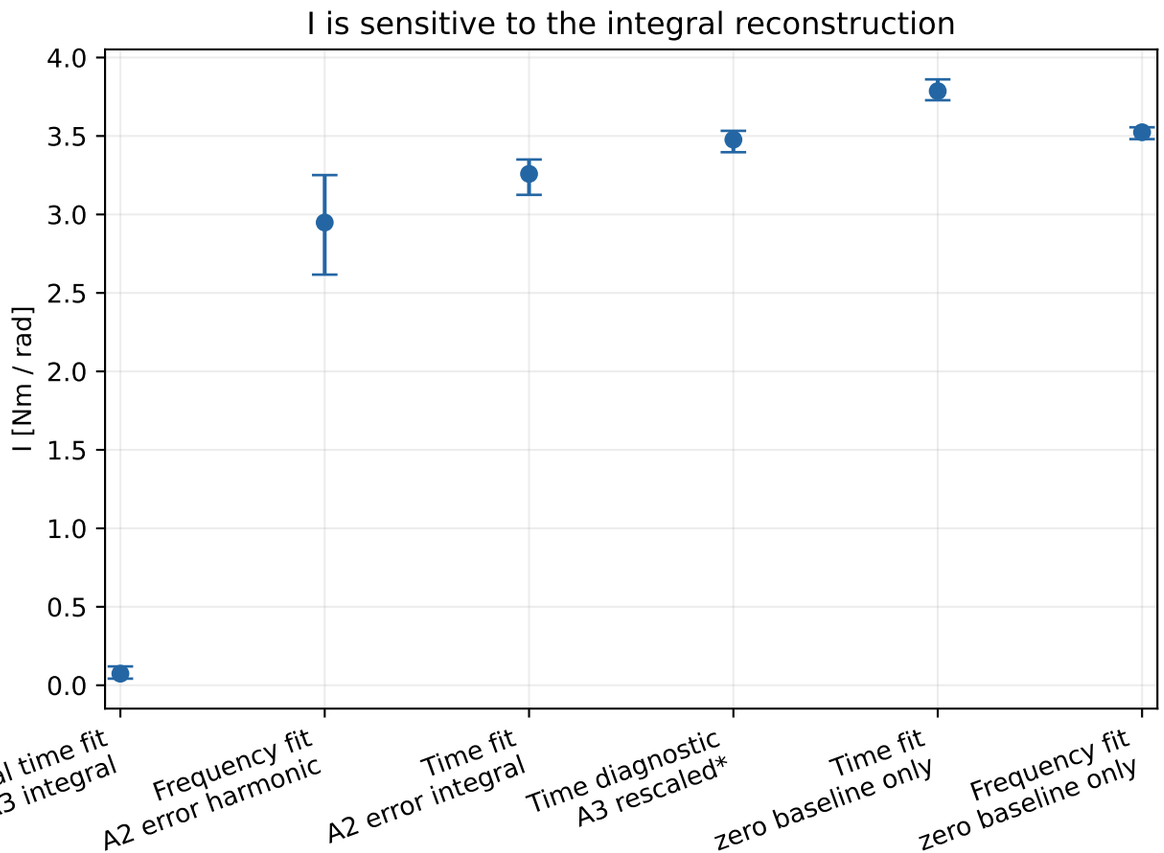
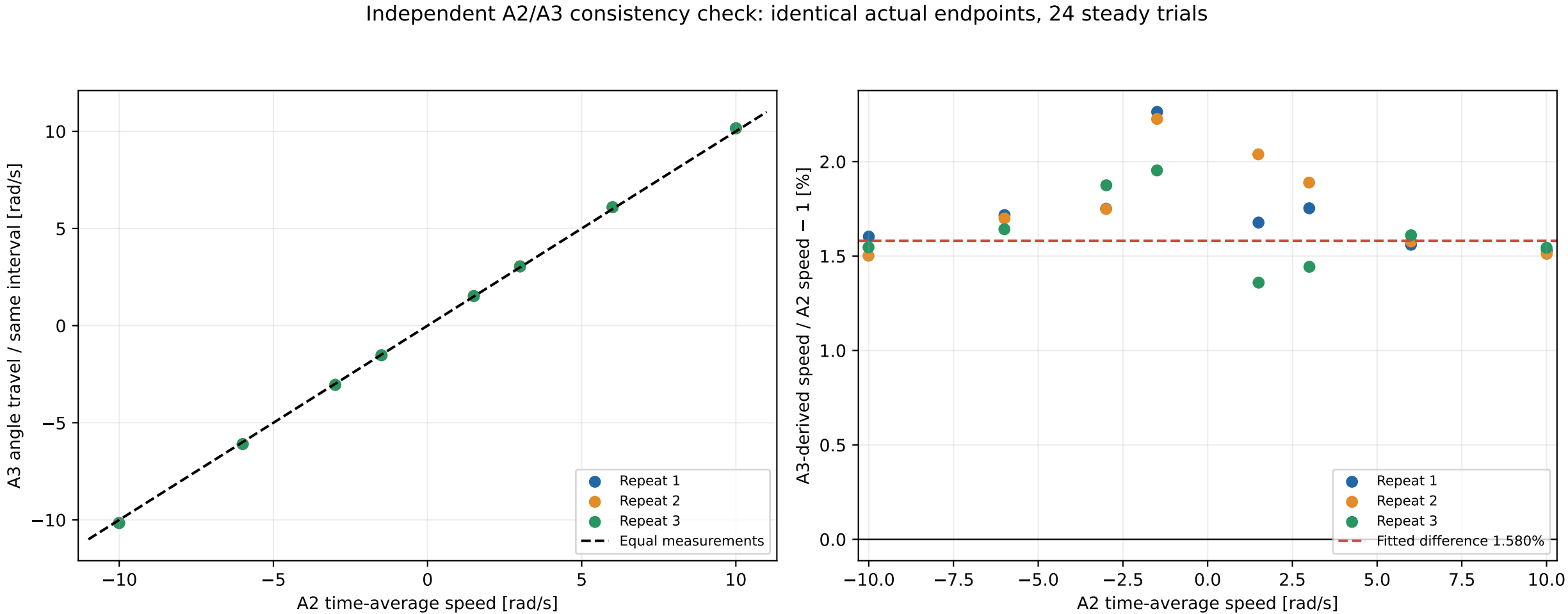
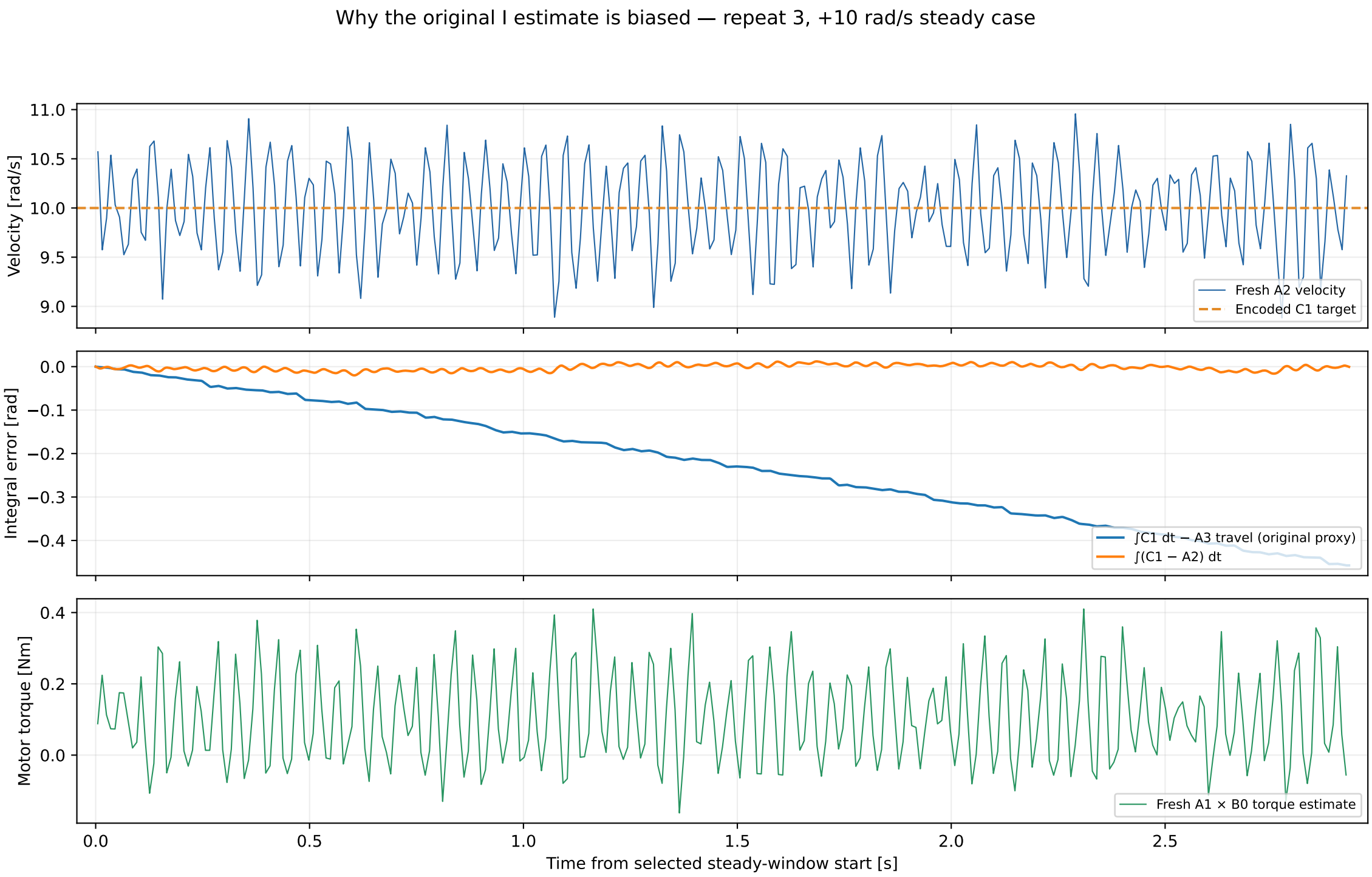
Rear C1 effective PI gains — unchanged firmware, airborne wheel

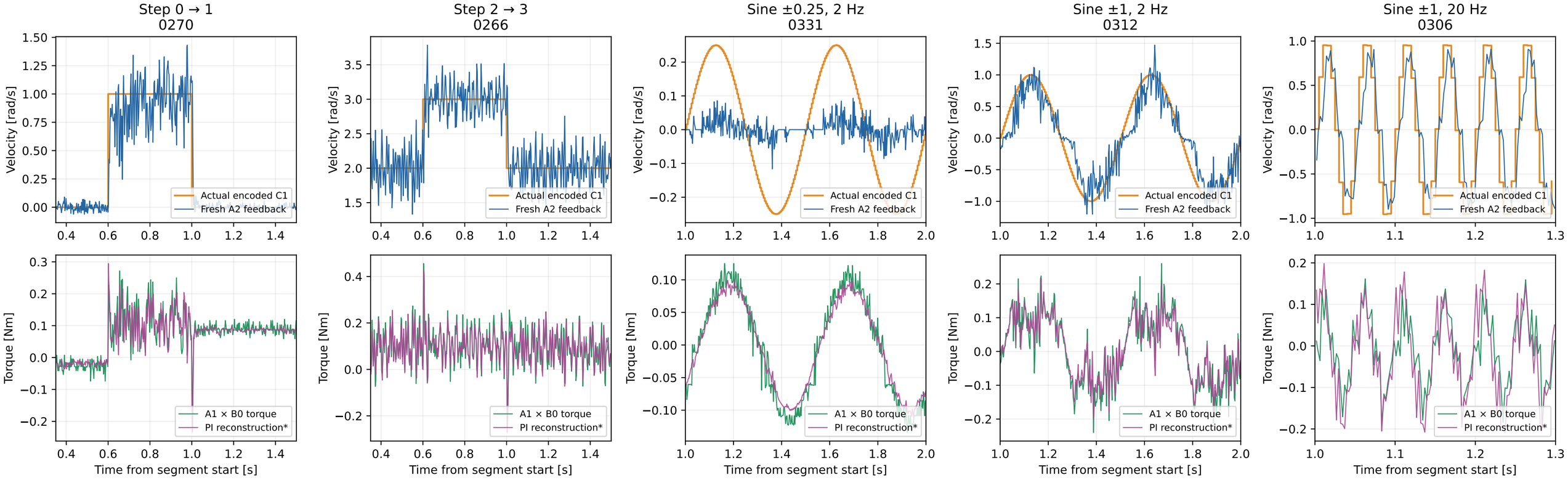


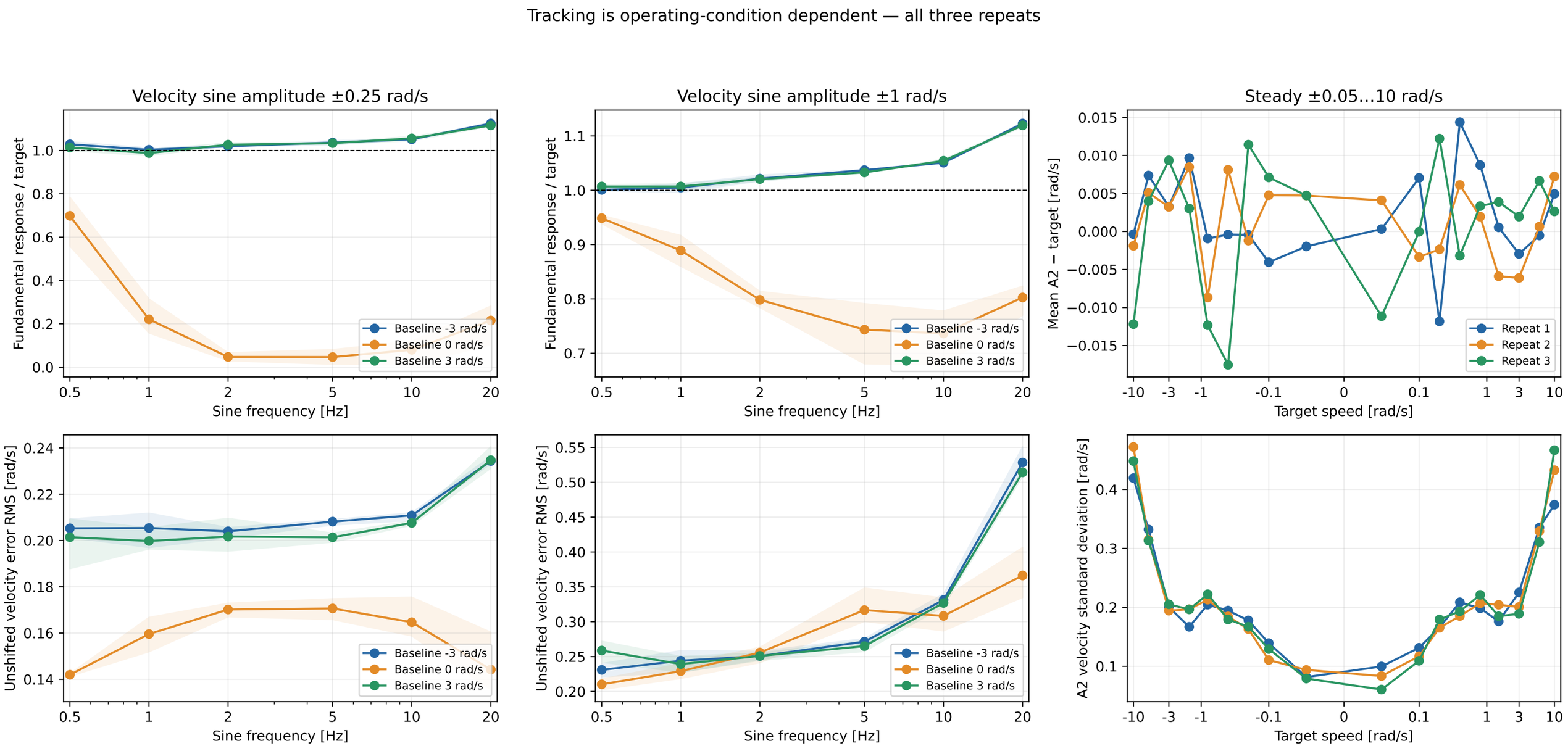


Bars: conditional 95% whole-trial bootstrap intervals; systematic uncertainty remains. *Diagnostic A3 scale fitted from repeat 1 only; no raw data changed.


Independent A2/A3 consistency check: identical actual endpoints, 24 steady trials
A3 angle travel / same interval [rad/s]
A2 time-average speed [rad/s]
Repeat 1
Repeat 2
Repeat 3
Equal measurements
A3-derived speed / A2 speed − 1 [%]
A2 time-average speed [rad/s]
Repeat 1
Repeat 2
Repeat 3
Fitted difference 1.580%
No time shift or physical calibration applied. Both directions and all three repeats show the discrepancy.


Why the original I estimate is biased — repeat 3, +10 rad/s steady case
Velocity [rad/s]
11.0
10.5
10.0
9.5
9.0
Fresh A2 velocity
Encoded C1 target
Integral error [rad]
0.0
−0.1
−0.2
−0.3
−0.4
∫C1 dt − A3 travel (original proxy)
∫(C1 − A2) dt
Motor torque [Nm]
0.4
0.2
0.0
Fresh A1 × B0 torque estimate
0.0
0.5
1.0
1.5
2.0
2.5
Time from selected steady-window start [s]
Integral origins set to zero at the analysis-window start only; hardware encoder origins are unchanged.


Measured C1 responses — repeat 3; near-zero reversals differ from rotating-wheel steps
Step 0 → 1
0270
Step 2 → 3
0266
Sine ±0.25, 2 Hz
0331
Sine ±1, 2 Hz
0312
Sine ±1, 20 Hz
0306
Velocity [rad/s]
Torque [Nm]
Actual encoded C1
Fresh A2 feedback
A1 × B0 torque
PI reconstruction*
Time from segment start [s]
*Uses observed error/integral, all-data fitted gains and a per-trial offset; this is not a forward simulation. Raw feedback timestamps are unshifted.


Tracking is operating-condition dependent — all three repeats
Velocity sine amplitude ±0.25 rad/s
Velocity sine amplitude ±1 rad/s
Steady ±0.05…10 rad/s
Fundamental response / target
Sine frequency [Hz]
Mean A2 − target [rad/s]
Target speed [rad/s]
Unshifted velocity error RMS [rad/s]
A2 velocity standard deviation [rad/s]
Baseline -3 rad/s
Baseline 0 rad/s
Baseline 3 rad/s
Repeat 1
Repeat 2
Repeat 3
Lines = mean of three trials; shading = trial range. Gains use actual held commands; RMS includes speed ripple and has no delay compensation.